Oscar A. Flores Gaitán

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Education

Ph.D. in Astronomy & Astrophysics — *Newark, DE*
Department of Physics & Astronomy, University of Delaware — Aug 2026 – Present
- Advisor: Prof. Sarah Dodson-Robinson

Honors Licentiate in Physics — *Guatemala City*
Department of Physics, Universidad del Valle de Guatemala — Jan 2022 – May 2026
- Honors: Magna Cum Laude
- Thesis: "*Decoding the Barnard's Star Planetary System: A Time Series Based Analysis of the Planetary Architecture*" (Advisor: Prof. Sarah Dodson-Robinson)

Research Interests

- Exoplanet detection, characterization, and planetary atmosphere modeling.
- Modeling and mitigating stellar activity in radial velocity and time-series data.
- Application of machine learning and advanced statistical methods to astronomical data.

Publications

First-Author

1. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V.. *"How many terrestrial planets orbit Barnard's Star?"*. [Manuscript in preparation, 2026].

Co-Authored & Contributor

2. Zhao, L.L. et al., **including Flores Gaitán, O.A.**. *"The Extreme Stellar-Signals Project IV"*. [Manuscript in preparation, 2026].

1. Dressing, C.D., Adams, D., Alecian, E. et al., **including Flores Gaitán, O.A.**. *"Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory"*. arXiv preprint (2026). arXiv:2608.11294.

Software

1. **Flores Gaitán, O.A.**, & Hopper, S.. *"orbcloud: 3D orbital probability clouds from simulated exoplanet posteriors"*. Zenodo (2026). DOI: 10.5281/zenodo.20935725.

Talks & Posters

6. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., Rubio-Herrera, E., *"How many terrestrial planets orbit Barnard's Star?"*, **Contributed Talk** at APS Mid-Atlantic Section Meeting 2025, Penn State Harrisburg, Middletown, PA, November 15, 2025. Link

5. **Flores Gaitán, O.A.**, *"Hunting for Exoplanets: Real Planets or Stellar Processes?"*, **Invited Talk** at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 25, 2025.

4. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, **Poster** at Physics Science Fair at Universidad del Valle de Guatemala's Physics Week, Guatemala City, October 24, 2025.

3. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, **Virtual Poster** at NASA's Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Johns Hopkins University, Washington, DC, July 28-31, 2025.

2. **Flores Gaitán, O.A.**, Dodson-Robinson, S., Ramirez Delgado, V., *"Disentangling Planetary and Stellar Signals in Barnard's Star"*, **Virtual Poster** at Sagan Summer Workshop, NASA

Exoplanet Science Institute, California Institute of Technology, Pasadena, CA, July 23, 2025. Link

1. **Flores Gaitán, O.A.**, *"Detecting and Characterizing Exoplanets with TESS"*, **Invited Talk** at Universidad del Valle de Guatemala's Astronomy Club, Guatemala City, May 9, 2025.

Research Experience

Undergraduate Visiting Researcher — *Newark, DE*
Department of Physics & Astronomy, University of Delaware — Jun 2025 – Jul 2026
Advisor: Prof. Sarah Dodson-Robinson

- Evaluated advanced statistical methods to distinguish genuine exoplanetary signals from stellar noise in high-precision radial velocity (RV) data.
- Implemented Gaussian Process (GP) regression and frequency-domain analysis to mitigate stellar variability and improve the reliability of RV detections.
- Analyzed 25 years of multi-instrument RV datasets to determine the planetary architecture of Barnard's Star, utilizing advanced statistical metrics for model validation.

Undergraduate Researcher — *Guatemala City*
Department of Physics, Universidad del Valle de Guatemala — Nov 2024 – May 2025
Advisor: Prof. Eduardo Rubio-Herrera

- Processed and analyzed photometric time-series data from the TESS mission to identify exoplanet candidates.
- Developed an algorithm to detect transits and extract orbital parameters from stellar light curves.

Industry & Professional Experience

Data & Research Analyst — *Guatemala City*
Vekst Consulting Group — Feb 2026 – May 2026

- Led scientific consulting and quantitative modeling projects, applying advanced statistical methods to financial and market data analysis.
- Transformed rigorous research into high-value strategic assets and technical documentation for the consulting sector.
- Designed, developed, and deployed the firm's official website, expanding technical and commercial visibility.

Academic Appointments & Teaching

Graduate Teaching Assistant — *Newark, DE*
Department of Physics & Astronomy, University of Delaware — Aug 2026 – Present

- Starting Fall 2026.

Undergraduate Teaching Assistant & Laboratory Assistant — *Guatemala City*
Physics Department, Universidad del Valle de Guatemala — Jan 2024 – May 2025

- Designed and implemented laboratory guides for undergraduate physics courses, assisted in the setup and supervision of laboratory sessions, and graded reports.
- Served as Teaching Assistant for 7 undergraduate physics courses, including Mechanics I and Physics I-III.

Academic Tutor — *Guatemala City*
Universidad del Valle de Guatemala — Jun 2022 – Present

- Provided 350+ hours of one-on-one tutoring for over 30 high school and college students, prepared personalized study materials, and tracked student progress.
- Subjects: Precalculus, Calculus I-III, Calculus for Business I-II, Differential Equations, Physics I-III.

Mentorship

University of Delaware Undergraduate Research Program *Newark, DE*
Primarily advised by Prof. Sarah Dodson-Robinson Jun 2026 – Present

- Student: Isabel Cambronero Jiménez (Universidad de Costa Rica)

Workshops & Summer Schools

Cohere Labs Machine Learning Summer School *Remote*
Cohere Labs Aug 2026

Code/Astro Workshop *Santa Cruz, CA*
University of California, Santa Cruz Jun 2026

Alpha-Cen Astronomy Summer School *Remote*
Alpha-Cen (Central American and Caribbean Astrophysics) Aug 2025

Workshop on Astrophysics of High Energies and Cosmology *Guatemala City*
Universidad Nacional Autónoma de México (UNAM) Nov 2024

Awards & Achievements

- **Summer Visiting Research Fellowship**, Department of Physics & Astronomy, University of Delaware — Selected out of 147 international applicants (2025).
- **3x Distinguished Student Award**, Universidad del Valle de Guatemala — Awarded for academic excellence and top GPA (2022 – 2024).
- **Top 10 Finalist**, Guatemalan National Mathematics Olympiad (2020).
- **Experimental Physics Record**, Universidad del Valle de Guatemala — Achieved the university's historically most accurate experimental measurement of gravitational acceleration (g) (2024).
- **2x National Scholar Champion**, High School National Soccer Championship.
- **Former Professional Soccer Player** (Goalkeeper), Guatemalan U-21 Professional Soccer League.
- **IRONMAN 70.3 Finisher** & Federated National Triathlete.

Technical Skills

Programming Languages: Python, LaTeX, R (Intermediate)

Data Analysis & Probabilistic Modeling: Statistical Inference, Gaussian Processes (GPs), Markov Chain Monte Carlo (MCMC), Bayesian Modeling, Time-Series Analysis

Machine Learning: Scikit-learn, XGBoost, Random Forests, Neural Networks, Autoencoders, K-Nearest Neighbors (KNN), Hyperparameter Tuning, Cross-Validation

Astronomy Tools: PyMC, celerite2, exoplanet, AstroPy, Lightkurve, Astroquery, ExoCTK, NWelch

Languages: Spanish (Native), English (C1), German (C1)

Outreach & Leadership

- Invited Panelist — *"Experiences of Internships Abroad for Aspiring Guatemalan Astronomers"*, CAFE 2026 (Workshop on Astronomical Sciences: Education and Training), Universidad Rafael Landívar, Guatemala City (Jul 2026).
- Science Fair Judge — Evaluated undergraduate physics and engineering research projects at Universidad del Valle de Guatemala's Science Fair, Guatemala City (Jun 2026).
- Workshop Developer & Instructor — Designed and taught *"How to Apply to Internships in the US"* for undergraduate physics students in Latin America (Feb 2026).
- Guest Lecturer & Teaching Assistant — Collaborated on Extragalactic Astronomy lecture for high school students, Astro Antigua, Antigua Guatemala (Dec 2025).
- Lead Organizer — Physics Week 2025, a week-long conference and workshop series featuring national and international speakers, Universidad del Valle de Guatemala, Guatemala City (Oct 2025).

- Workshop Developer & Instructor — Designed and taught *"How to Write an Academic CV"* for first-year physics students, Universidad del Valle de Guatemala, Guatemala City (Oct 2025).
- Active Member — Physics Students Association (AEF-UVG), organizing public lectures and science outreach events (2025).
- Author & Content Creator — Authored undergraduate Physics laboratory guide and created educational laboratory videos at Universidad del Valle de Guatemala (2024). Link
- Founder & Manager — *Countries Kingdom*, educational geography and linguistics outreach project with over **15,000 followers** and **5,000,000+ total views** (2021 – 2023). Link

Media

3. **Actualidad UVG**, *"Oscar Flores Gaitán demonstrates the reach of Physics in the search for exoplanets (In Spanish)"*, **Feature Article** at Universidad del Valle de Guatemala, Guatemala City, 2026. Link
2. **Ciencia UVG Podcast**, *"A UVG Physicist in Delaware (In Spanish)"*, **Podcast Episode** at Universidad del Valle de Guatemala, YouTube (Video) / Spotify (Audio), 2026.
1. **Educational Video Tutorial**, *"Measuring the gravitational acceleration g with a physical pendulum (In Spanish)"*, **YouTube Video** at Universidad del Valle de Guatemala, Guatemala City, 2024. Link